IDENTITY ACCESS MANAGEMENT PROJECT (ACTIVE DIRECTORY)

Project Objectives

The project's primary focus was installing and hardening Active Directory on a newly provisioned Windows Server. After establishing a private NAT Network, the server was promoted to a Domain Controller, creating the wyzlo.com. The administrative structure was then organized with OUs, Groups, and Users, followed by the successful integration of a client endpoint. The hardening phase involved creating and linking restrictive Group Policy Objects (GPOs) to limit user and computer capabilities like disabling shutdown and command prompt access.

Organizational Units (OUs)

Active Directory Users and Computers

Wyzlo.com

OU: USA

└── Group: IT

└── Users: Mark.IT

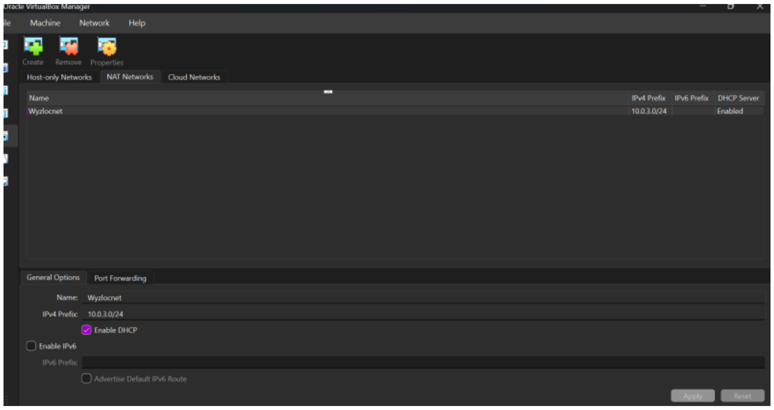
OU: CANADA

└── Group: SALES

└── Users: Singh.SALES

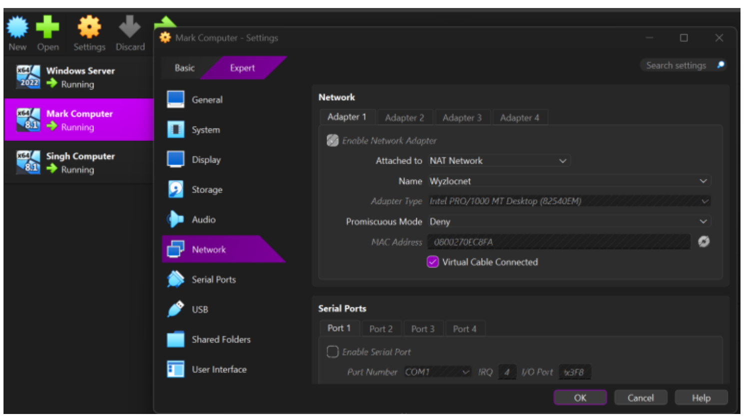
After securing the VM access, I will create the NAT Network for Subnet Isolation

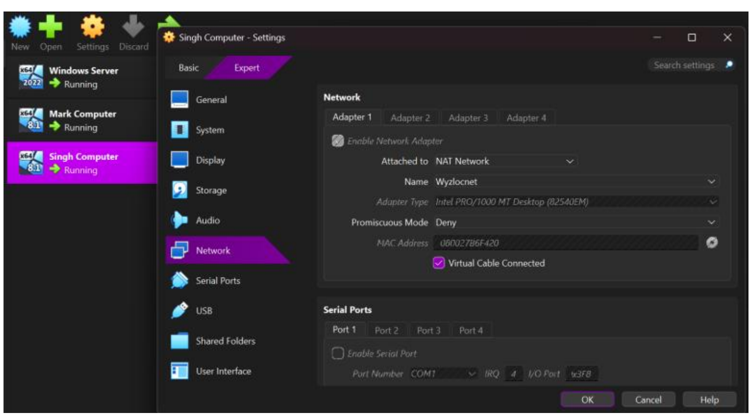
I accessed the VirtualBox network settings by navigating to the main menu (Tools) and selecting Network. Inside the network options, I choose the NAT Network option, then clicked create. The new network was named Wyzlocnet. The Dynamic Host Configuration Protocol (DHCP) service was explicitly enabled to automatically assign IP addresses to the connecting virtual machines, simplifying the network setup.

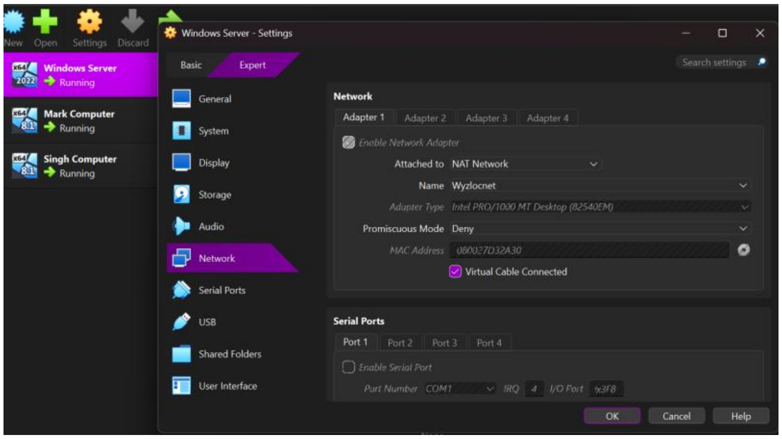


Next step is to put the clients on the network just created

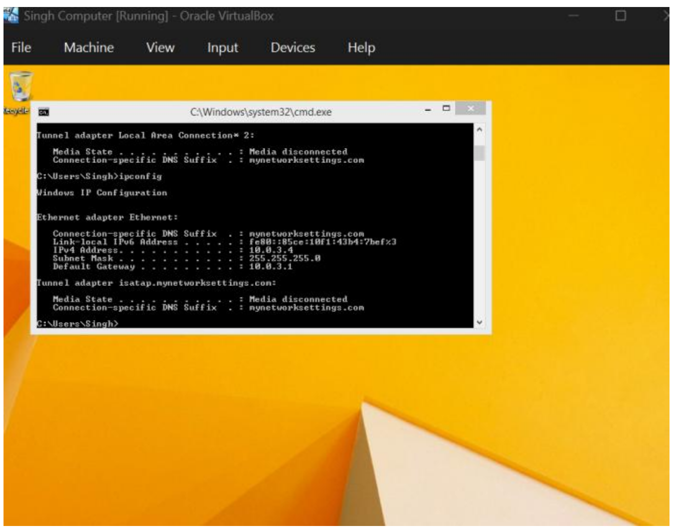
Click settings > under Expert tab select Network > Attached to drop down choose NAT Network > Make sure Name is the right on > press ok



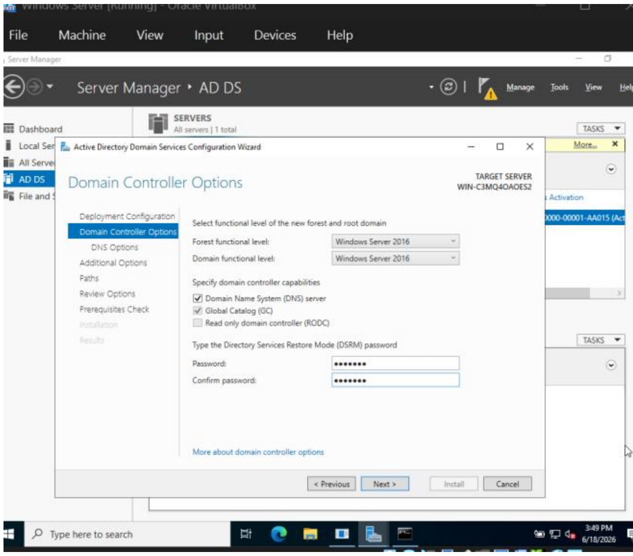


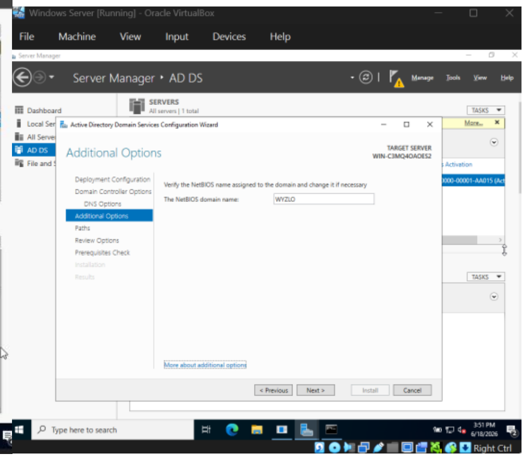


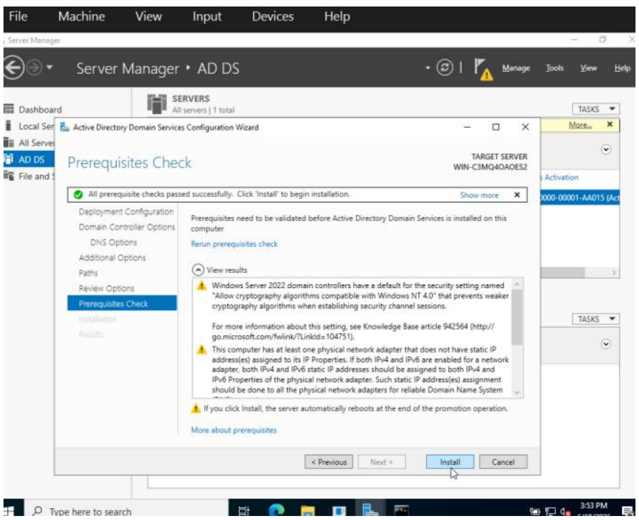
Confirming by typing ipconfig, when confirmed successful network connectivity to the subnet gateway, this concludes the foundational network setup



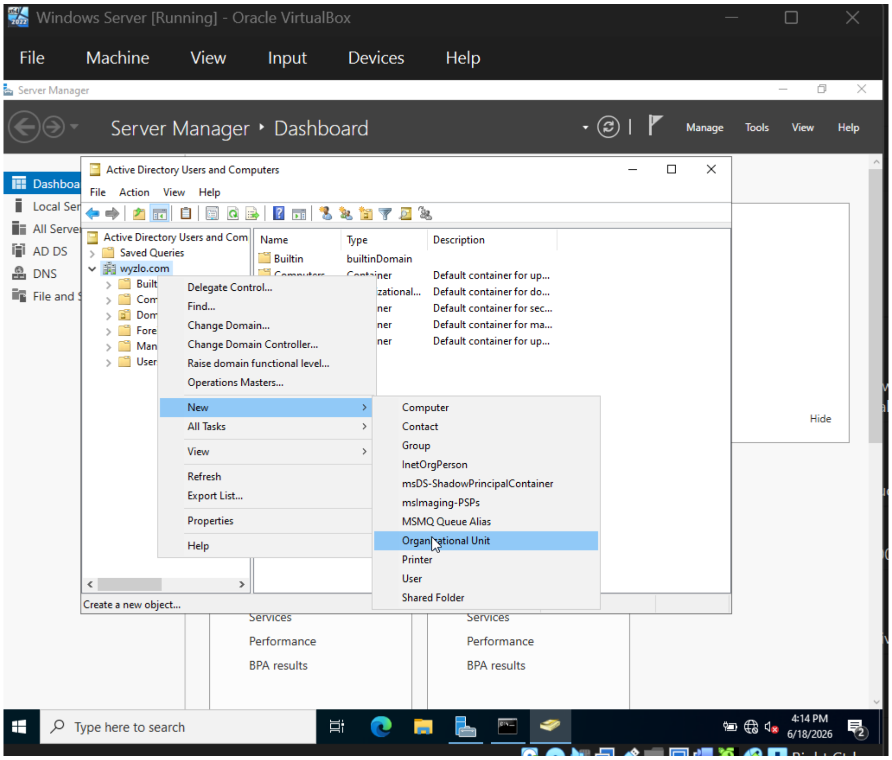
Promoting server to a domain controller



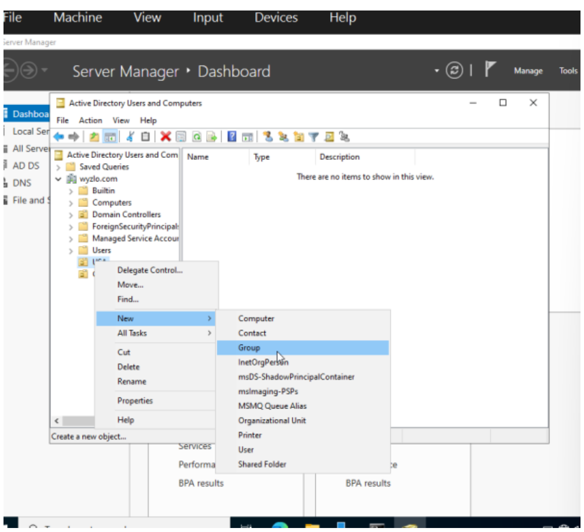


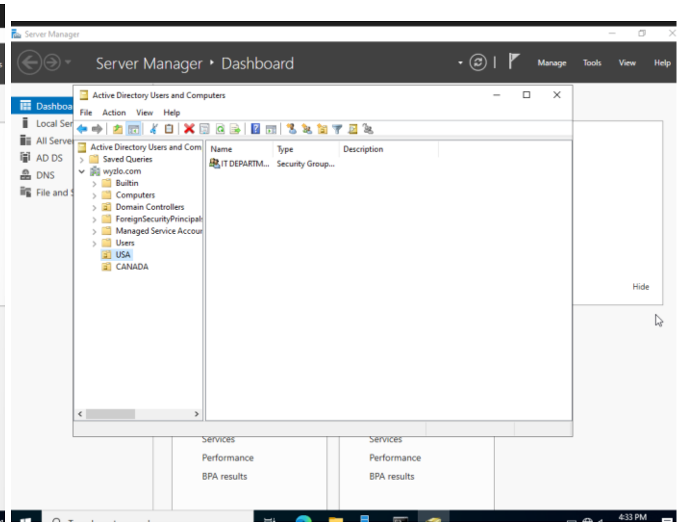


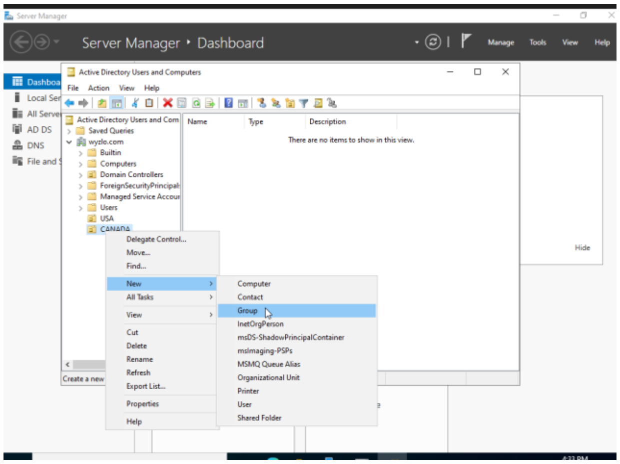
Creating Organizational Units (OUs): tools > active directory users and computers > right click domain forest > click new > find Organizational Unit

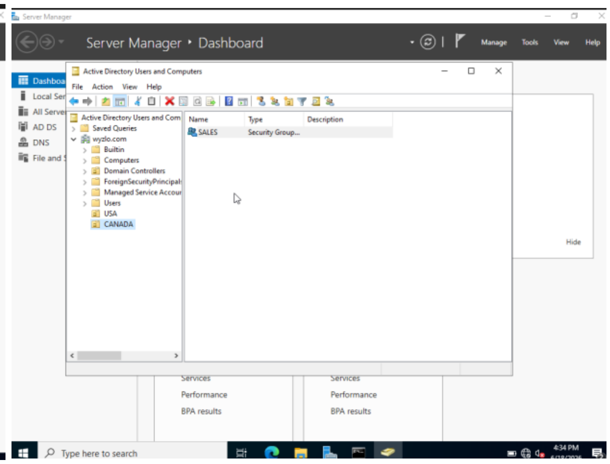


Creating Groups: right click on OU > click new > find Group

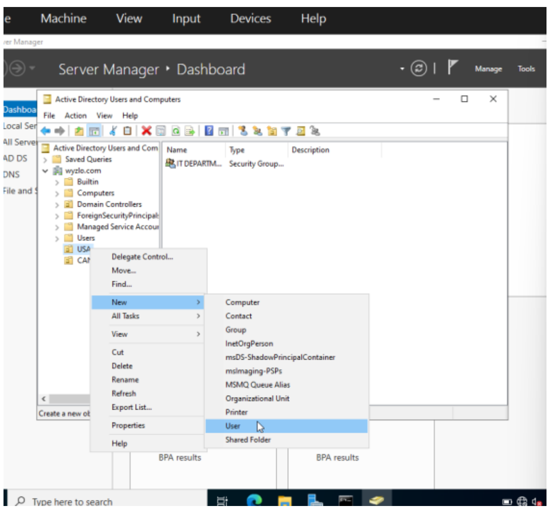


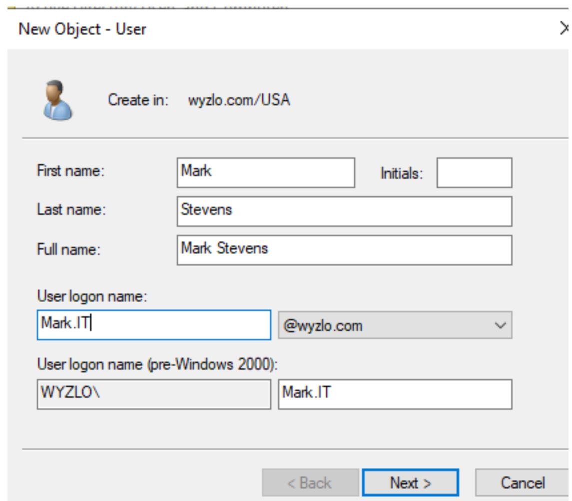


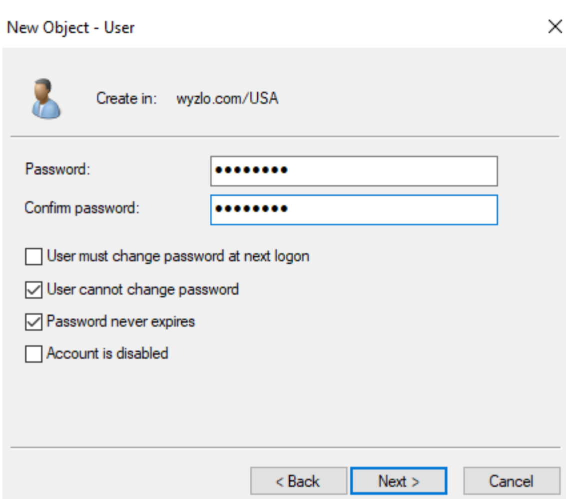


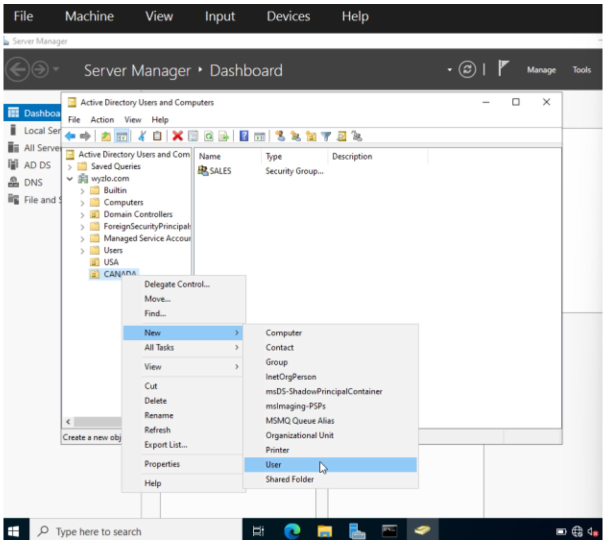


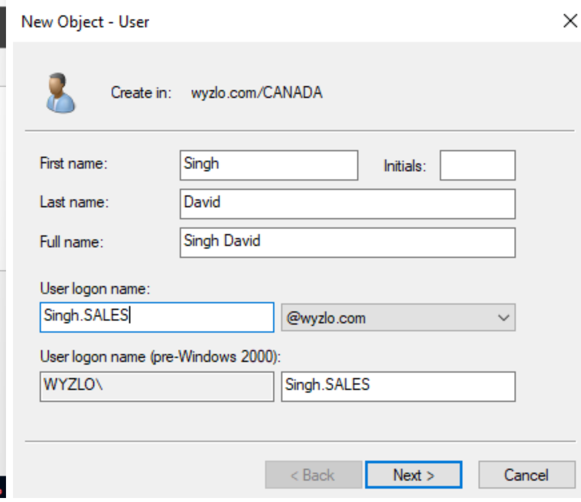
Creating Users: right click on OU > click new > find User

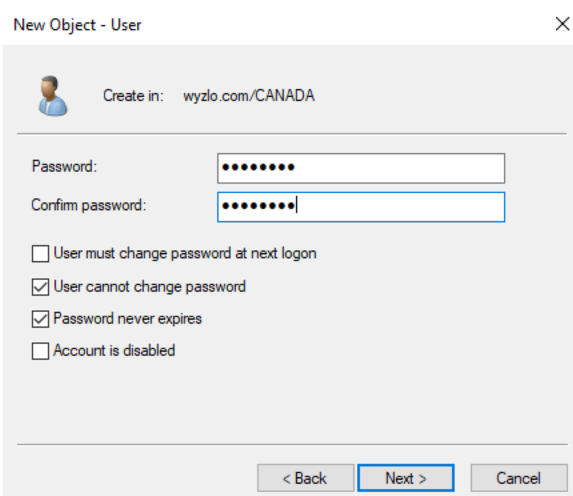




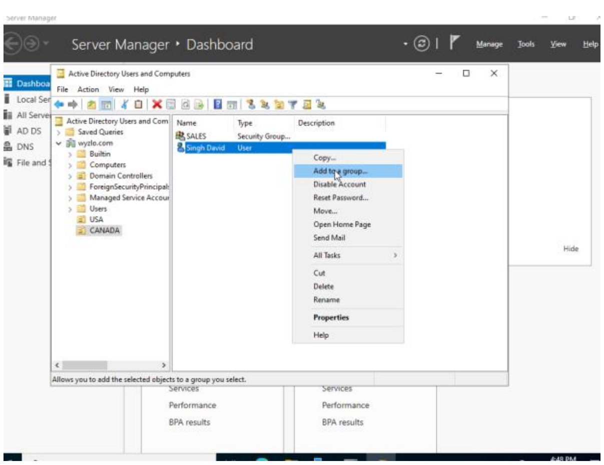


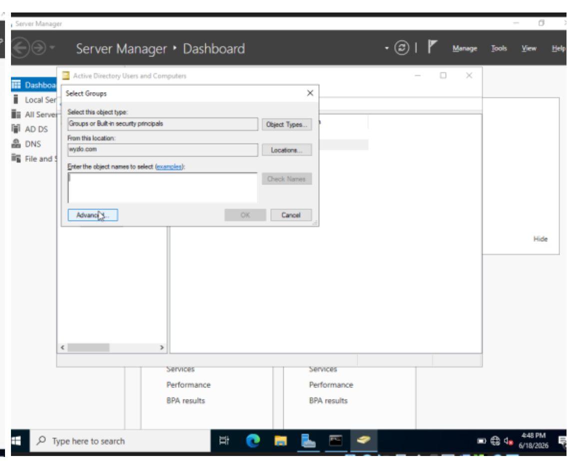


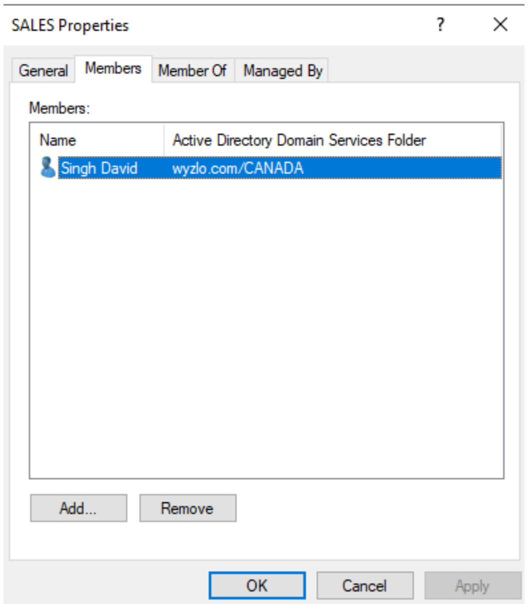


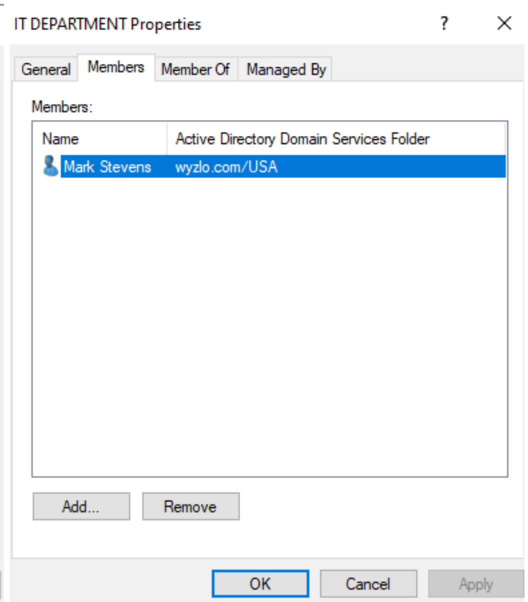


Right click user > add to a group > click Advanced > press find now > find Name > Press OK

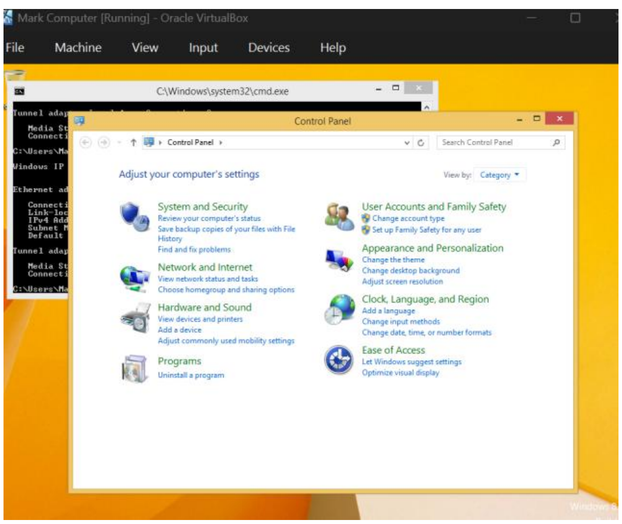




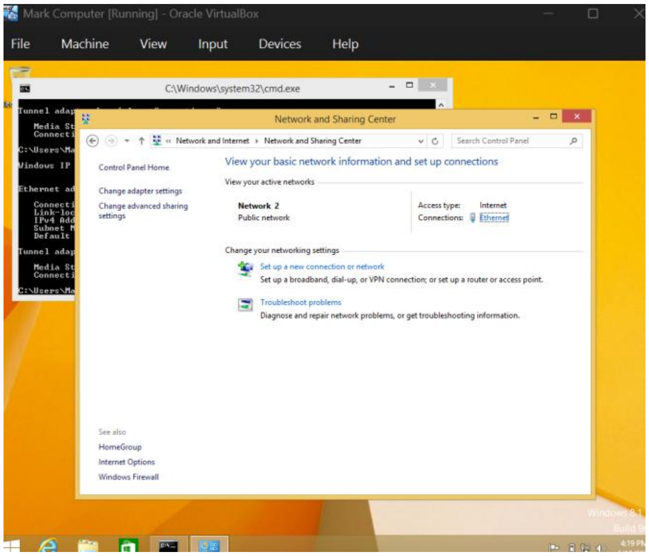




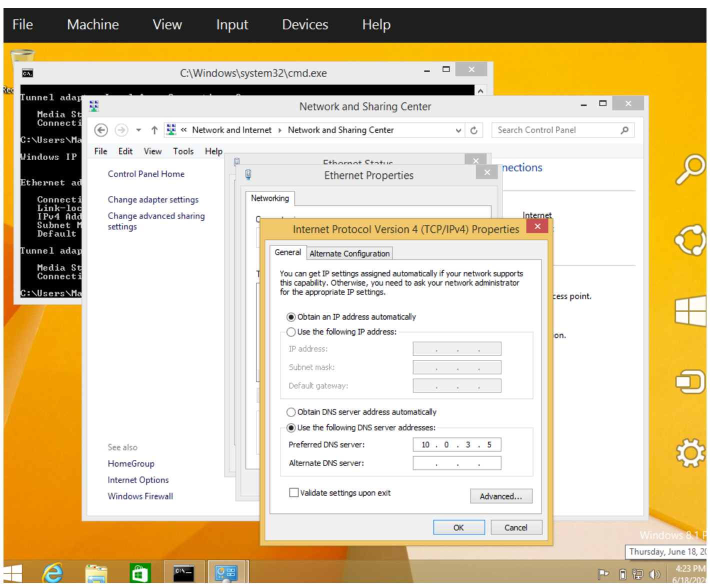
DNS of user PC to server IP: go to control panel >

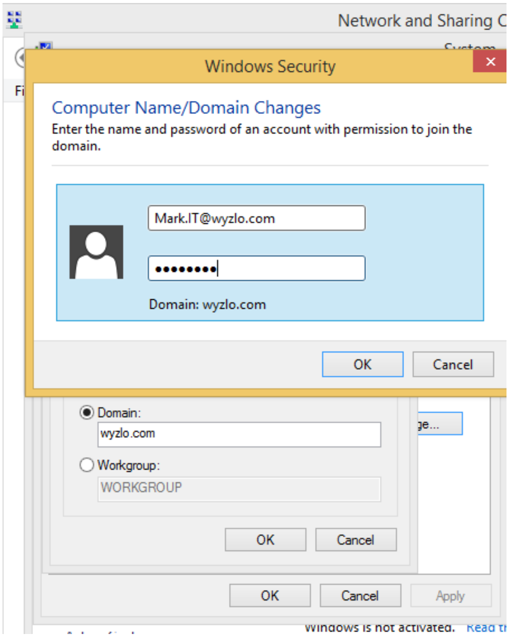


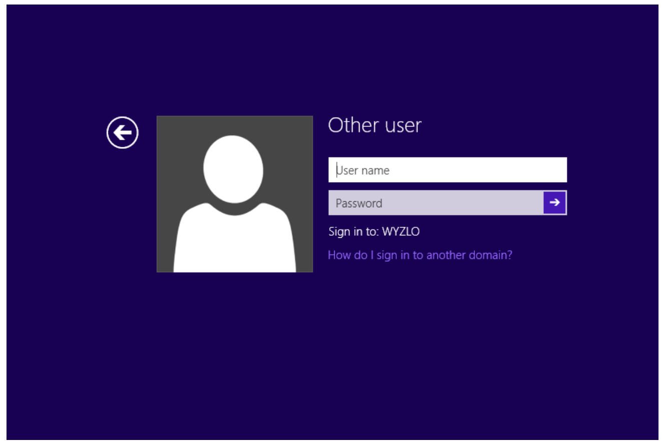
Network and Internet > Network and Sharing Center



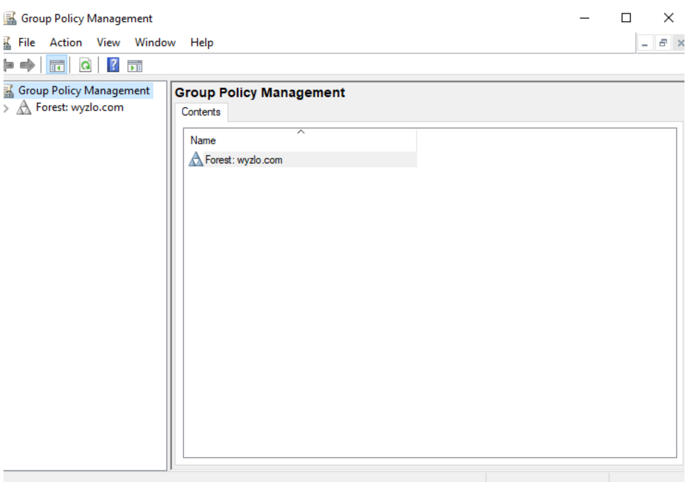
Find Ethernet > click properties > find IPv4 > click properties >







Group Policy: Win+R > gpmc.msc



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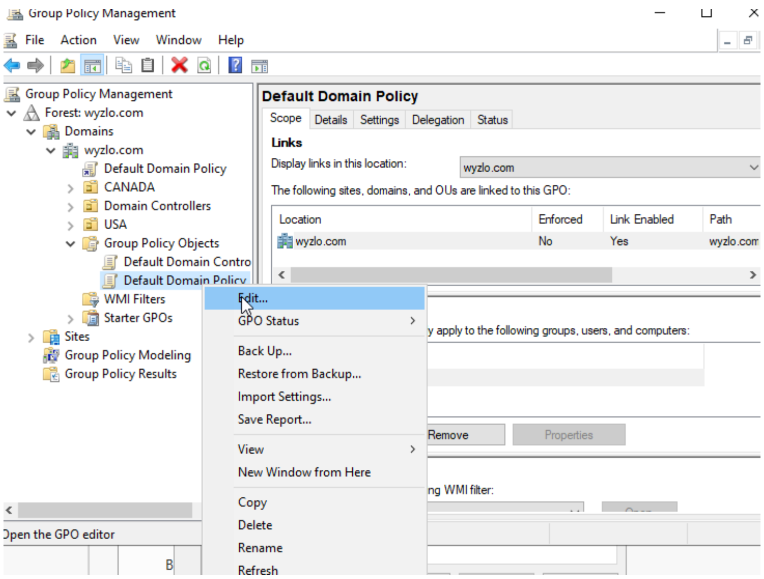
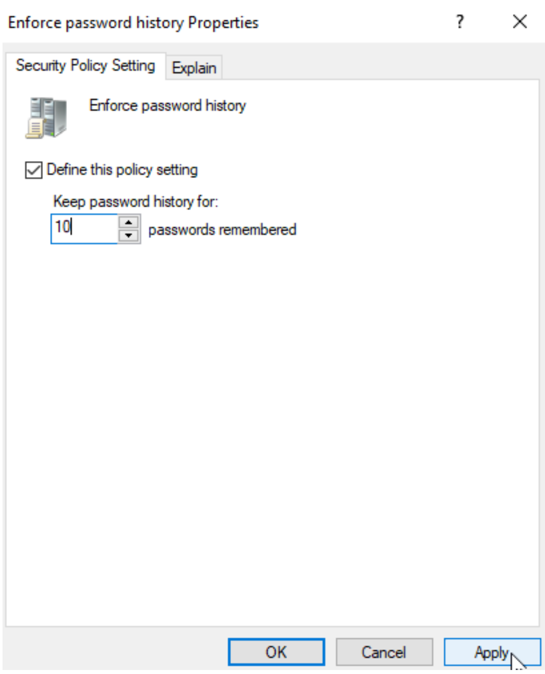
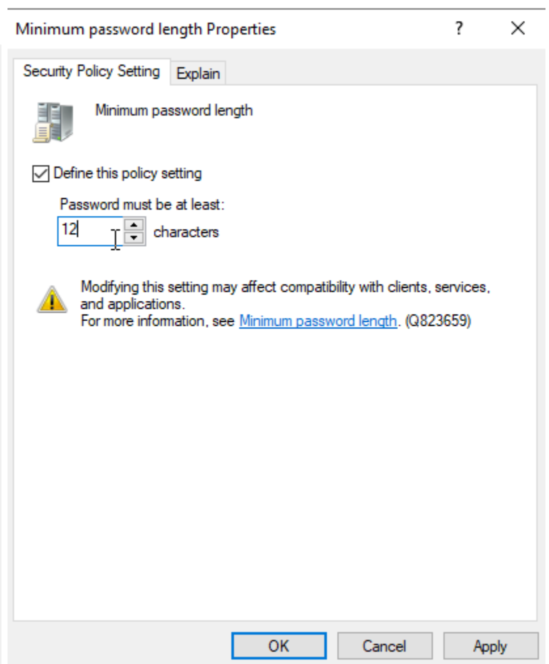


Group Policy Management
File Action View Window Help
Group Policy Management
Forest: wyzlo.com
Domains
wyzlo.com
Default Domain Policy
CANADA
Domain Controllers
USA
Group Policy Objects
Default Domain Contro
Default Domain Policy
WMI Filters
Starter GPOs
Sites
Group Policy Modeling
Group Policy Results
Default Domain Policy
Scope Details Settings Delegation Status
Links
Display links in this location:
wyzlo.com
The following sites, domains, and OUs are linked to this GPO:
Location Enforced Link Enabled Path
wyzlo.com No Yes wyzlo.com
Edit...
GPO Status
Back Up...
Restore from Backup...
Import Settings...
Save Report...
View
New Window from Here
Copy
Delete
Rename
Refresh
y apply to the following groups, users, and computers:
Remove
Properties
ng WMI filter:
Open the GPO editor

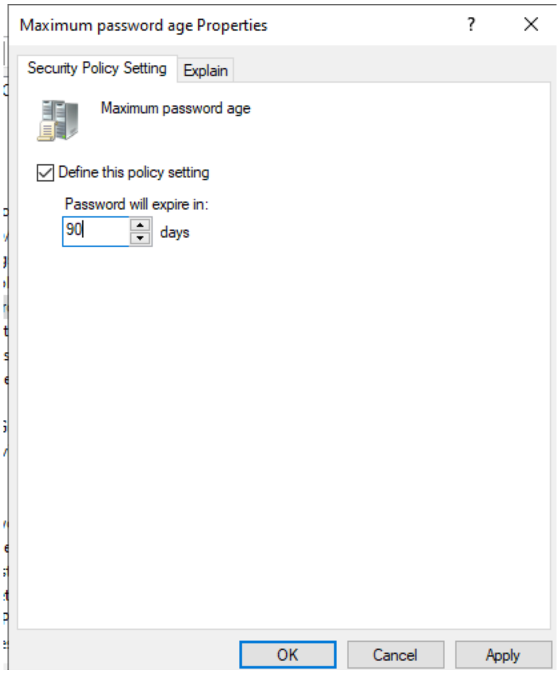
>

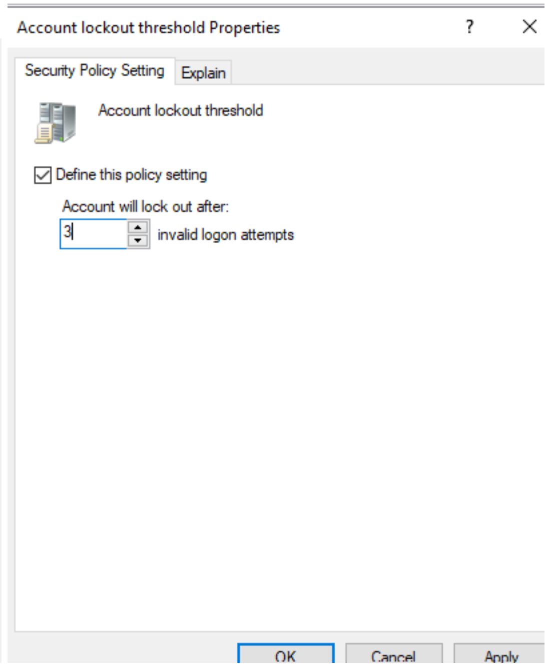


Enforce password history Properties
Security Policy Setting
Explain
Enforce password history
Define this policy setting
Keep password history for:
10
passwords remembered
OK
Cancel
Apply



Minimum password length Properties
Security Policy Setting
Explain
Minimum password length
Define this policy setting
Password must be at least:
12
characters
Modifying this setting may affect compatibility with clients, services, and applications.
For more information, see Minimum password length. (Q823659)
OK
Cancel
Apply





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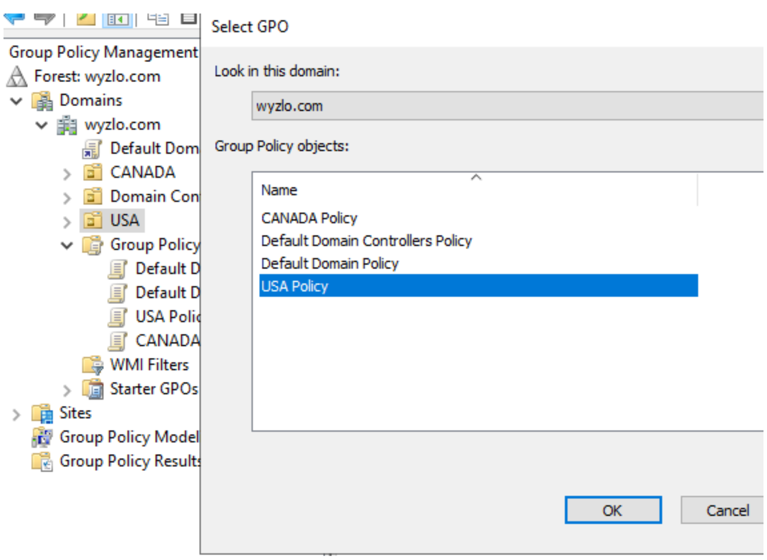
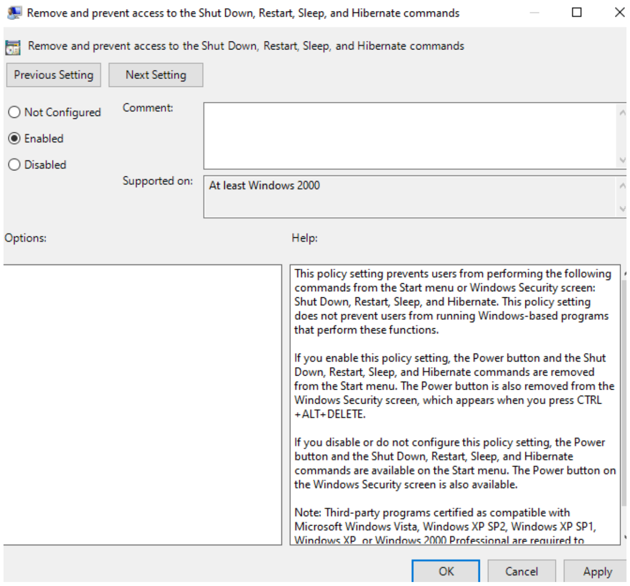
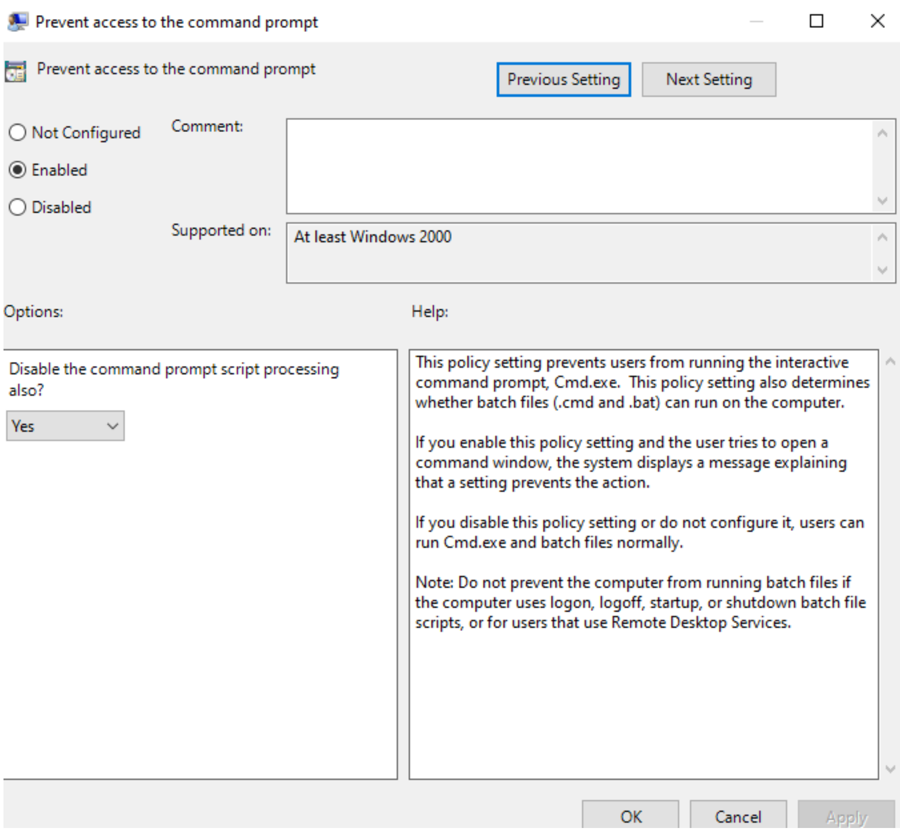
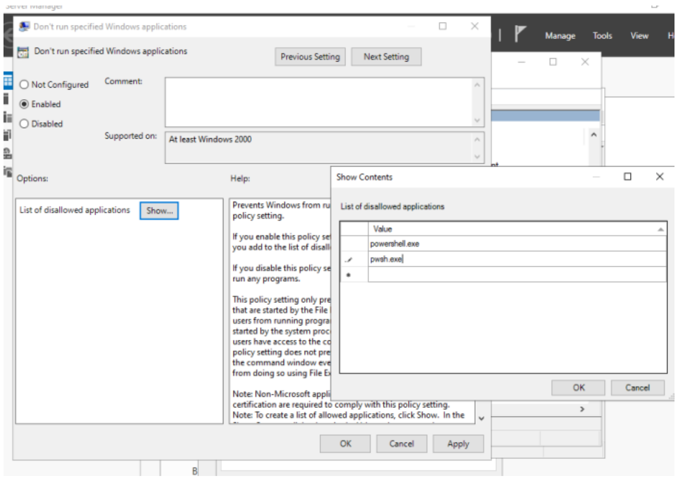


disallowing applications

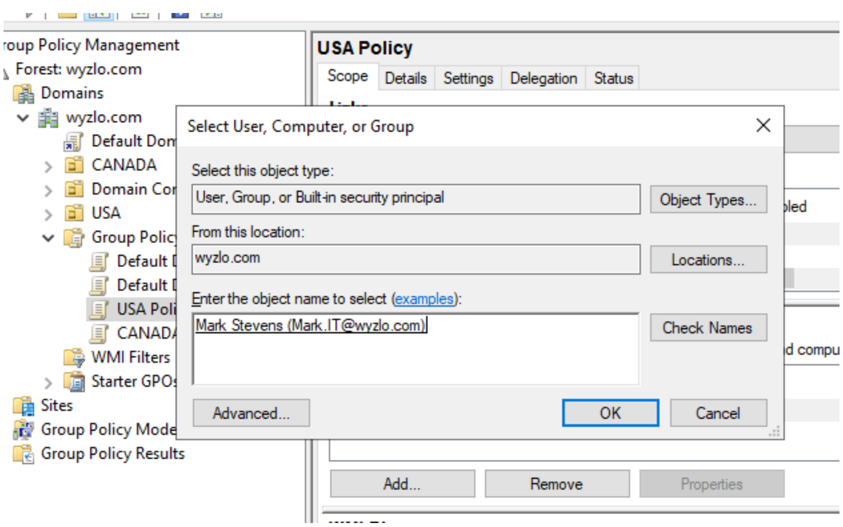

Remove and prevent access to the Shut Down, Restart, Sleep, and Hibernate commands
Remove and prevent access to the Shut Down, Restart, Sleep, and Hibernate commands
Previous Setting
Next Setting
Not Configured
Enabled
Disabled
Comment:
Supported on:
At least Windows 2000
Options:
Help:
This policy setting prevents users from performing the following commands from the Start menu or Windows Security screen: Shut Down, Restart, Sleep, and Hibernate. This policy setting does not prevent users from running Windows-based programs that perform these functions.
If you enable this policy setting, the Power button and the Shut Down, Restart, Sleep, and Hibernate commands are removed from the Start menu. The Power button is also removed from the Windows Security screen, which appears when you press CTRL +ALT+DELETE.
If you disable or do not configure this policy setting, the Power button and the Shut Down, Restart, Sleep, and Hibernate commands are available on the Start menu. The Power button on the Windows Security screen is also available.
Note: Third-party programs certified as compatible with Microsoft Windows Vista, Windows XP SP2, Windows XP SP1,
OK
Cancel
Apply


Prevent access to the command prompt
Prevent access to the command prompt
Previous Setting
Next Setting
Not Configured
Enabled
Disabled
Comment:
Supported on:
At least Windows 2000
Options:
Help:
Disable the command prompt script processing also?
Yes
This policy setting prevents users from running the interactive command prompt, Cmd.exe. This policy setting also determines whether batch files (.cmd and .bat) can run on the computer.
If you enable this policy setting and the user tries to open a command window, the system displays a message explaining that a setting prevents the action.
If you disable this policy setting or do not configure it, users can run Cmd.exe and batch files normally.
Note: Do not prevent the computer from running batch files if the computer uses logon, logoff, startup, or shutdown batch file scripts, or for users that use Remote Desktop Services.
OK
Cancel
Apply

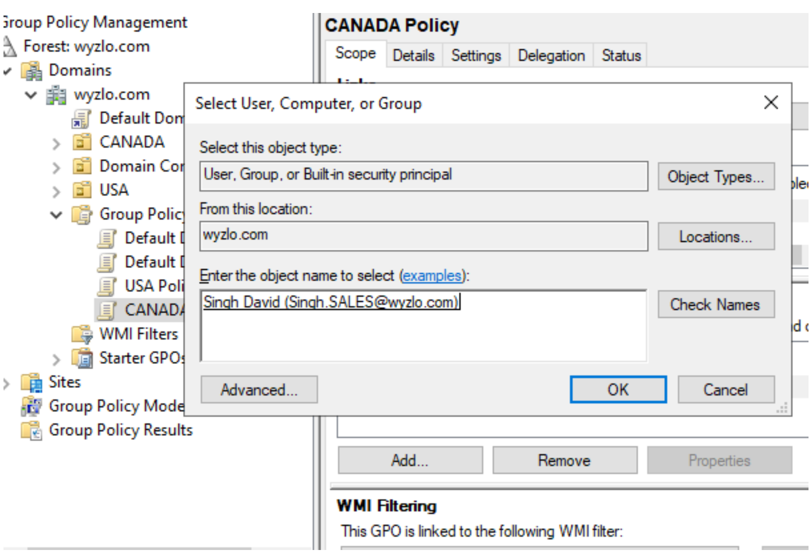


Don't run specified Windows applications
Previous Setting
Next Setting
Not Configured
Enabled
Disabled
Comment:
Supported on:
At least Windows 2000
Options:
Help:
List of disallowed applications
Show...
Show Contents
List of disallowed applications
Value
powershell.exe
pwsh.exe
OK
Cancel
Apply
Manage
Tools
View

>

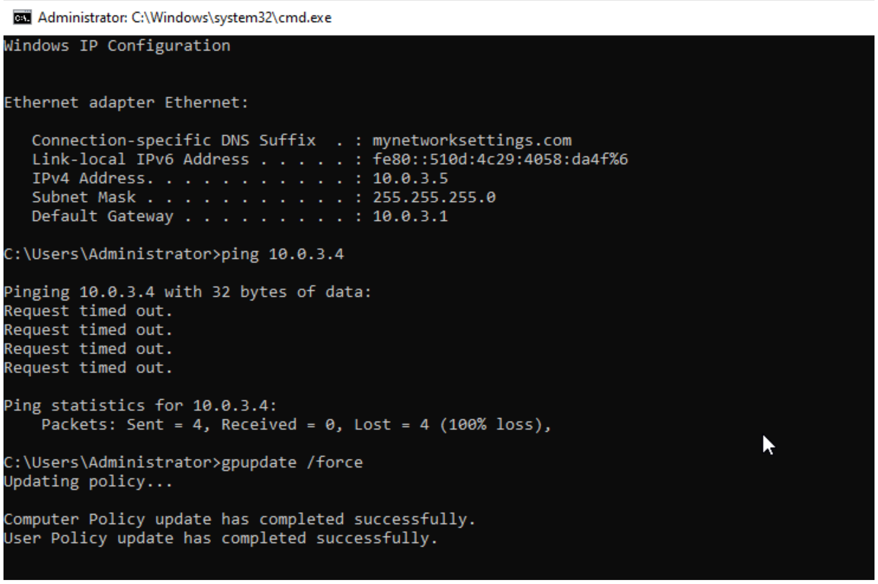


Forest: wyzlo.com
Domains
wyzlo.com
CANADA
USA
WMI Filters
Sites
Group Policy Results
USA Policy
Scope
Details
Settings
Delegation
Status
Select User, Computer, or Group
Select this object type:
User, Group, or Built-in security principal
Object Types...
From this location:
wyzlo.com
Locations...
Enter the object name to select (examples):
Mark Stevens (Mark.IT@wyzlo.com)
Check Names
Advanced...
OK
Cancel
Add...
Remove
Properties



Forest: wyzlo.com
Domains
wyzlo.com
CANADA
USA
WMI Filters
Sites
Group Policy Results
CANADA Policy
Scope
Details
Settings
Delegation
Status
Select User, Computer, or Group
Select this object type:
User, Group, or Built-in security principal
Object Types...
From this location:
wyzlo.com
Locations...
Enter the object name to select (examples):
Singh David (Singh.SALES@wyzlo.com)
Check Names
Advanced...
OK
Cancel
Add...
Remove
Properties
WMI Filtering
This GPO is linked to the following WMI filter:

Updating policy:



End Result Summary:

The project delivered a secure, centrally managed domain environment with unified authentication and role-based access through security groups. Hardening policies were successfully enforced via Group Policy, blocking shutdown and command-line access on the client machine. Verification with *gpupdate /force* confirmed that the domain's security controls are active and functioning as intended.